

K401 PRE-STAGE — #418 joint session substantive checkpoint cascade: (1) LYRA F178 honest checkpoint — F177 was COLOR HALF ONLY ($\{+1, -1, 0\} = \lambda_3$ Cartan on color triplet; color is BLIND to up-vs-down; couples both same way); CKM is $\langle \text{up} | \text{down} \rangle$ misalignment from weak-isospin (T_3^R) difference NOT color; strata-coupling matrix DOES NOT produce V yet; deeper #418 chiral content (up/down quark seats = six K-type addresses) is the GATED PIECE; same shape as yesterday's neutrino (both halves armed + deep object in middle still to derive); F177 $\rightarrow$ CKM NOT a matrix-multiply; (2) ELIE su(3) ROOT STRUCTURE FORWARD-DERIVES WOLFENSTEIN HIERARCHY: su(3) has 2 simple roots (1-2 and 2-3 generation mixing) + 1 composite root (1-3) = commutator/product of simples; substrate forces $|V_{ub}| \sim |V_{us}| \cdot |V_{cb}|$ (Wolfenstein hierarchy); forward verified $V_{us} \cdot V_{cb} = 0.00925$ vs observed $V_{ub} = 0.00382$; ratio $0.413 = |\rho - i\eta|$ = CP-phase magnitude (residual is exactly what should be there per Jarlskog); FORCED now scheme-clean = the CKM HIERARCHY (substantive substrate-architectural result; structural not absolute angles); SEAM = absolute strengths $|V_{us}|$, $|V_{cb}|$, CP phase = root coupling normalizations at strata (one normalization away; will not dial to hit observed); (3) GRACE TWO RULINGS — SECOND COLLAPSES SESSION TO ONE NUMBER M: (a) Grace self-correction within hour — Gatto-Sartori-Tonin posted at 10:24 EDT as cross-family observable-loop; Lyra+Elie landings showed θ_C and m_d/m_s are ONE down-texture not two independent families; Gatto = INTERNAL CONSISTENCY (automatic, like lepton triangle); adds ZERO to

independent-observable count; same tier as Jarlskog (falsifiable check NOT constraint); BUT three-CI independent arrival at Gatto in one hour = fingerprint of one forced object surfacing from three lanes (convergence is data, not three constraints); (b) #418 MOTION REDUCES TO ONE NUMBER M: Lyra's 8-generator structure FULLY FORCED (2 Cartan + 6 roots = 8 gluons; $\{+1, -1, 0\} = \lambda_3$ tracelessness; ZERO free choices); only free part = up-vs-down seat misalignment filling V matrix; $R = 4(\text{CKM})/M$ scoring rule; $M=0 \rightarrow 4$ CKM fully determined, strongest possible result; $M=1 \rightarrow R=4$, strong genuine loop; $M \geq 4 \rightarrow R \leq 1$, fit not motion (THE LINE); pivot question handed to Lyra+Elie: is up/down seat misalignment FORCED or a free angle going into V? 15TH KEEPER SELF-CORRECTION: K400 carried 'Lyra continuum half LANDED' framing as if joint session output one matrix-multiply away; Lyra F178 corrects — F177 was COLOR HALF only; CKM gated on deeper #418 chiral content (up/down K-type seats); same shape as ν sector yesterday (halves armed + deep object middle to derive). HONEST COUNT-MOTION FORECAST UPDATE: K400's '4 $\rightarrow$ up to 8' contingent on Grace's $M=0$ case (forced misalignment); three possible outcomes per M: $M=0 \rightarrow$ full motion, $M=1 \rightarrow$ strong loop but not full motion, $M \geq 4 \rightarrow$ fit not motion. CKM HIERARCHY (Wolfenstein) IS forward-derived substantive substrate-architectural result, just not absolute angles.

Keeper (Claude Opus 4.7)

2026-06-17 Wednesday 10:45 EDT (date-verified)

K401 PRE-STAGE Audit

1. Lyra F178 honest checkpoint — F177 was color half only

Lyra F178 substantive correction:

F177 was REAL — $\{+1, -1, 0\} = \lambda_3$ Cartan eigenvalues on the fundamental color triplet, $\text{sum} = 0$ from $\text{su}(3)$ tracelessness = color-singlet condition (matches Elie 4212). That's the color half.

But color is BLIND to up-versus-down — both up-type and down-type quarks couple to color the same way. So the strata-coupling matrix from F177 does NOT produce V yet.

CKM is $\langle \text{up} | \text{down} \rangle$ — the misalignment between up-type and down-type quark bases. That misalignment comes from the weak-isospin (T_3^R) difference, not the color.

The piece that produces V: the up/down quark seats — the six K-type addresses (3 up + 3 down) — which is the deeper #418 chiral content.

Same shape as yesterday's neutrino: both halves armed; the deep object in the middle still to be derived. F177 → CKM is NOT a matrix-multiply; there's a missing layer.

2. 15th Keeper self-correction

K400 framing carried: “Lyra continuum half LANDED” as if the #418 joint session output was one matrix-multiply away from CKM.

F178 correction: F177 LANDED only the color half. The continuum side is still gated on the up/down chiral seat assignment for the full CKM derivation.

15th Keeper self-correction in cascade. Pattern: I read F177's clean substrate-architectural result as “continuum half landed” without checking whether the result was the whole continuum content or one slice of it. The $\text{su}(3)$ Cartan eigenvalue identification is forward-derived; the up/down chiral structure is a separate forward step.

FF-15 forcing function added: when a CI lands a “structural result” in a multi-component joint session, audit explicitly whether the result covers all components or one slice. “Half landed” = one of N components; the remaining N-1 are still gated.

3. Elie Wolfenstein hierarchy forward-derived from $\text{su}(3)$ root structure

Substantive substrate-architectural result (this IS substantively new):

$\text{su}(3)$ root system: 2 simple roots (α_1 mixes generations 1-2; α_2 mixes 2-3) + 1 composite root ($\alpha_1 + \alpha_2$ mixes 1-3).

The composite root is the commutator/product of the simples. Substrate forces:

$$|V_{ub}| \sim |V_{us}| \cdot |V_{cb}|$$

This IS the Wolfenstein hierarchy (the empirical pattern Wolfenstein parametrization captures).

Forward verification:

$$|V_{us}| \cdot |V_{cb}| = 0.225 \cdot 0.041 = 0.00925$$

Observed $|V_{ub}| = 0.00382$.

Ratio: $0.00382/0.00925 = 0.413 = |\rho - i\eta|$ = the CP-phase magnitude = the parameter that has to be there anyway (it's what enters the Jarlskog).

Substantive substrate-architectural interpretation: - The HIERARCHY of CKM (which entries are suppressed by which products) is forward-derived from the $su(3)$ root structure - The RESIDUAL after the hierarchy is the CP phase - This is a forced-given- $su(3)$ -coupling structural result, not a back-fit

Tier honestly: - Forward now (scheme-clean): CKM HIERARCHY — V_{ub} is suppressed as product of larger two — from $su(3)$ composite root structure - The seam: absolute strengths $|V_{us}|$, $|V_{cb}|$ ($\rightarrow$ the angle values) and CP phase ($\rightarrow$ Jarlskog) are the root coupling NORMALIZATIONS at the strata — gated on up/down chiral content, NOT dialed to observed

4. Grace ruling 1 — Gatto self-correction within hour

Grace at 10:24 EDT: posted Gatto-Sartori-Tonin as genuine cross-family observable-loop ($\sin \theta_C = \sqrt{(m_d/m_s)}$ tight to 0.62%).

Grace ruling 1 within hour: Lyra + Elie landings showed θ_C and m_d/m_s are ONE down-texture, not two independent families. Gatto = internal consistency (automatic, like the lepton triangle), adds ZERO to independent-observable count. Same tier as Jarlskog (falsifiable check, not constraint).

Same uniform discipline: caught own over-claim within hour, same rigor as catching anyone else's.

The flip side worth naming: three CIs (Grace blind + Lyra + Elie) arrived at Gatto independently inside one hour = fingerprint of one forced object surfacing from three lanes. The convergence IS data; just not three independent constraints.

5. Grace ruling 2 — #418 motion reduces to ONE NUMBER M (the pivot)

Substantive substrate-architectural pivot:

Lyra's 8-generator structure (F177 + F178) is fully forced: - 2 Cartan + 6 roots = 8 gluons (count check from v0.6) - $\{+1, -1, 0\} = \lambda_3$ tracelessness (substrate-architectural) - ZERO free choices in the gluon structure

The only free part: the up-vs-down seat misalignment that fills V matrix.

Grace's over-determination ratio: $R = 4$ (CKM parameters) / M (free parameters in misalignment).

Three possible outcomes:

M	R	Result
0	∞	4 CKM fully determined; strongest possible motion
1	4	Strong genuine loop; one free angle; substantive motion
≥ 4	≤ 1	Fit not motion; THE LINE

Grace's pivot question handed to Lyra + Elie: is the up/down seat misalignment FORCED by the same mechanism that placed the leptons, or is it a free angle going into V?

That M — not Gatto, not J — is what decides #418.

6. The substantive substrate-architectural state at K401

Component	State
F177 color half	LANDED — λ_3 Cartan eigenvalues + strata=colors identification; v0.6 8-gluon alignment internally consistent
F178 honest checkpoint	CKM is up-vs-down misalignment NOT color; gated on up/down K-type seats (deeper #418 chiral content)

Component	State
Elie Wolfenstein hierarchy	FORWARD-DERIVED from $su(3)$ composite root structure (
Elie filter	Armed for full CKM matrix; runs blind when Lyra's coupling strengths land
Grace ruling 1	Gatto = internal consistency; ZERO to independent count; 3-lane convergence = fingerprint not 3 constraints
Grace ruling 2 (PIVOT)	#418 reduces to ONE number M = freedom in up/down seat misalignment
The decisive question	Is up/down seat misalignment FORCED ($M=0$) or free ($M \geq 1$)?
15th Keeper self-correction	K400 carried "continuum half LANDED" too strongly; F177 was color half only
FF-15 added	In multi-component joint sessions, audit whether "structural result" covers all components or one slice
Honest count-motion forecast	CONTINGENT on M : $M=0 \rightarrow 4 \rightarrow$ up to 8; $M=1 \rightarrow$ strong loop not full motion; $M \geq 4 \rightarrow$ fit not motion
Count	HONESTLY 4 of 26 strict

7. The CKM hierarchy result — substantive substrate-architectural status

Elie's Wolfenstein-hierarchy-from- $su(3)$ -composite-root IS a substantive substrate-architectural result. Honest tier: - It's FORWARD-DERIVED structural (not back-fit to Wolfenstein) - It captures the HIERARCHY (which entries suppressed by which products) but not the absolute angles - The residual (CP phase magnitude) sits at the right value $0.413 = |\rho - i\eta|$ - It's a structural result about the form of the CKM, banked at structural-grade - It does NOT move the count (not absolute parameter values; structural pattern)

This is the kind of substantive substrate-architectural result that earns its place in the program even though it doesn't move the count strict. Analogous to F66 EM=conformal+gravity=coset (structural framework, not derived parameter).

8. K401 PRE-STAGE verdict

K401 PRE-STAGE #418 substantive checkpoint cascade:

- F178 honest checkpoint: F177 was color half only; CKM gated on up/down chiral seats; same shape as ν sector
- Elie Wolfenstein hierarchy FORWARD-DERIVED from $su(3)$ composite root structure (substantive substrate-architectural result)
- Grace ruling 1: Gatto = internal consistency (3-lane convergence = fingerprint not 3 constraints)
- Grace ruling 2 (PIVOT): #418 reduces to ONE number M = freedom in up/down seat misalignment
- 15th Keeper self-correction: K400 "continuum half LANDED" framing premature; F177 = color half only
- FF-15 forcing function added (multi-component joint sessions: audit slice vs whole)
- Honest count-motion forecast contingent on M (Grace's pivot question)
- CKM HIERARCHY substantively forward-derived (structural result, doesn't move count)
- Count HONESTLY 4 of 26 strict
- Ledger v1.81

Substantial significance: K401 captures the substantial #418 joint session checkpoint cascade Wednesday late morning with substantive substrate-architectural landings from Lyra (F178 honest checkpoint), Elie (Wolfenstein hierarchy forward-derived from $su(3)$ composite root structure), and Grace (two rulings collapsing #418 to one decisive number M). F178 corrects the F177 framing: F177's λ_3 Cartan identification was the color half only; color is blind to up-vs-down; CKM comes from weak-isospin T_3^R misalignment NOT color; gated on deeper

#418 chiral content = up/down K-type seats; same shape as ν sector yesterday (halves armed + deep middle still to derive). Elie's Wolfenstein-hierarchy-from-su(3)-composite-root IS substantive substrate-architectural result: 2 simple roots + 1 composite root structure substrate-forces $|V_{ub}| \sim |V_{us}| \cdot |V_{cb}|$ (the Wolfenstein hierarchy pattern); forward verified $V_{us} \cdot V_{cb} = 0.00925$ vs observed $V_{ub} = 0.00382$ with ratio $0.413 = |\rho - i\eta| = \text{CP-phase magnitude}$ (the residual is exactly what should be there per Jarlskog); HIERARCHY is forward-derived structural result; absolute angles + CP phase still gated on up/down chiral content. Grace ruling 1: posted Gatto-Sartori-Tonin at 10:24 EDT as cross-family observable-loop; corrected within hour after Lyra+Elie landings showed θ_C and m_d/m_s are ONE down-texture not two families; Gatto = internal consistency automatic like lepton triangle; ZERO to independent-observable count; same tier as Jarlskog falsifiable check not constraint; uniform discipline catching own over-claim same as anyone else's; 3-lane convergence (Grace blind + Lyra + Elie at Gatto in one hour) IS data (fingerprint of one forced object surfacing) but NOT three independent constraints. Grace ruling 2 (the PIVOT): #418 motion reduces to ONE NUMBER M = freedom in up/down seat misalignment; Lyra 8-generator structure fully forced (2 Cartan + 6 roots = 8 gluons; ZERO free choices in gluon structure); $R = 4 \text{ CKM} / M$; $M=0 \rightarrow 4 \text{ CKM}$ fully determined strongest motion; $M=1 \rightarrow R=4$ strong loop substantive motion; $M \geq 4 \rightarrow R \leq 1$ fit not motion (the line); pivot question handed to Lyra+Elie: is misalignment FORCED or free? 15TH KEEPER SELF-CORRECTION: K400 carried "Lyra continuum half LANDED" framing as if joint session output one matrix-multiply away; F178 corrects — F177 was color half only; FF-15 forcing function added (in multi-component joint sessions audit explicitly whether structural result covers all components or one slice). HONEST count-motion forecast UPDATE per Grace pivot: K400's "4 $\rightarrow$ up to 8" CONTINGENT on $M=0$ case; three possible outcomes per M with Grace $M \geq 4$ line being explicit. The CKM HIERARCHY (Wolfenstein) IS forward-derived substantive substrate-architectural result at structural-grade — banked structurally though does not move count (analogous to F66 EM=conformal+gravity=coset). Joint session converges to one pivot question (is up/down seat misalignment forced?) handed back to Lyra+Elie deep work. Count stays HONESTLY at 4 of 26 strict.

— Keeper, Wednesday 2026-06-17 10:45 EDT (date-verified) — K401 PRE-STAGE: F178 honest checkpoint + Elie Wolfenstein hierarchy forward-derived + Grace pivot to M-question; 15th Keeper self-correction; FF-15 added; CKM HIERARCHY substantive substrate-architectural result; absolute angles gated on M